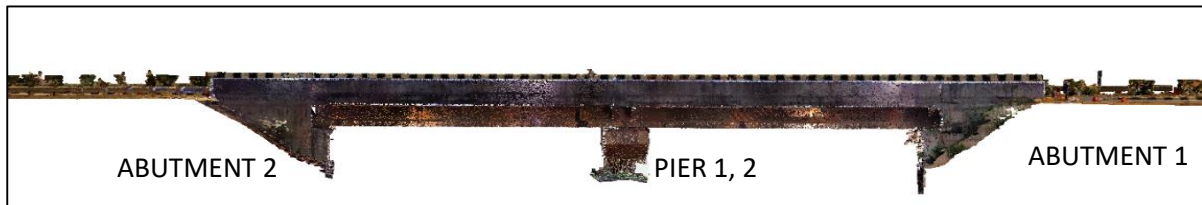


# **LIDAR Inspection Report of MNB ch.946+200**

**Bridge No:** MNB - 947/1

**Bridge Chainage:** 946+200 (CKRP)

**Latitude/Longitude:** 25.310834, 74.918060



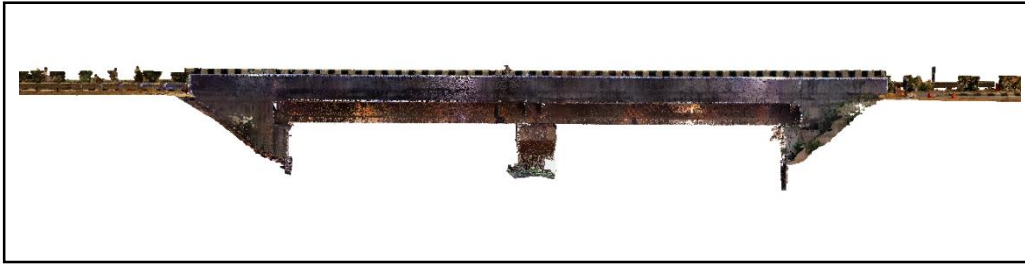
➤ **Bridge Details**

- **Bridge No:** 947/1
- **Chainage:** 946+200
- **Structure type:** MNB

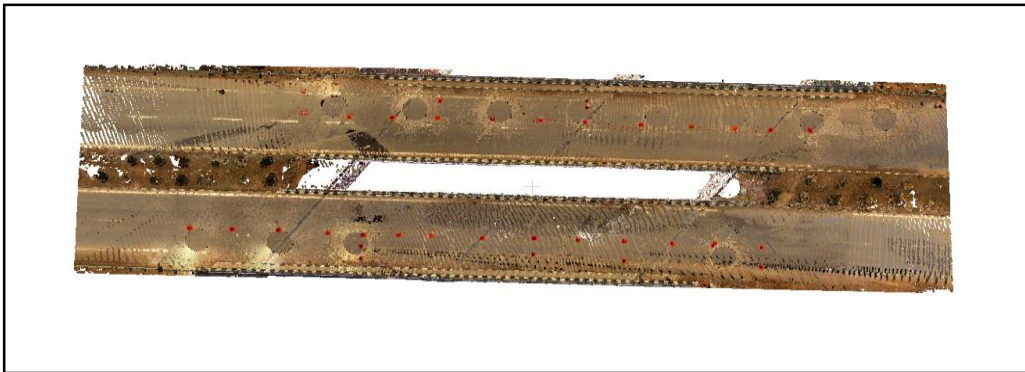
➤ **Project Details**

- **CKRP:** Chittorgarh to kota
- **State:** Rajasthan NH 27
- **Zone:** RJ-06

## LiDAR View:



Side View



Top View



Bottom View



Front View

## Structural Measurements

### 1. Height

| Measurement                | Value  |
|----------------------------|--------|
| Pier bottom to Slab bottom | 4.26 m |
| Abutment bottom to Road    | 6.60m  |

### 2. Length

|     | Abutment to Abutment | Span 1  | Span 2  |
|-----|----------------------|---------|---------|
| LHS | 30.56 m              | 14.29 m | 14.28 m |
| RHS | 30.56 m              | 14.29 m | 14.28 m |

Deck Width (LHS): 10.10 m

Deck Width (RHS): 10.10 m

### 3. Width Between Piers to Abutment

| Measurement | Abutment 1 to Pier 1 | Pier 1 to Abutment 2 |
|-------------|----------------------|----------------------|
| Value       | 14.29 m              | 14.28 m              |

## Technical Scope of LiDAR Inspection

| Category                                      | Details                                                                                                                                                    |
|-----------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>1. Accuracy Setup and Scan Planning</b>    |                                                                                                                                                            |
| Equipment Deployed                            | <b>Trimble X9</b> – High-performance terrestrial laser scanner with automatic calibration, IP65-rated rugged build, and in-field registration capability   |
| Point Cloud Density                           | Configured for high-density capture with sub-centimeter resolution suitable for fine structural detailing                                                  |
| Measurement Accuracy                          | <b>±1 mm</b> at close range (<60 m); optimal for bridge spans, elevated members, and fascia with scan range up to <b>150 meters</b>                        |
| Coverage Strategy                             | Comprehensive planning to prioritize critical structural components including abutments, piers, bearings, and deck; ensured overlap to minimize occlusion  |
| <b>2. Field Setup and Scanning Approach</b>   |                                                                                                                                                            |
| Site Configuration                            | Multiple scan positions arranged for 360° coverage including underside views; referenced globally using <b>GNSS/RTK and Ground Control Points (GCPs)</b>   |
| Execution Environment                         | Scans executed during low-traffic, clear weather conditions to minimize external noise and vibration-related scan distortion                               |
| Safety Measures                               | Operational safety ensured by advance site clearance, barricading, and coordination with local authorities for traffic management                          |
| <b>3. Data Resolution and Quality Control</b> |                                                                                                                                                            |
| Resolution Application                        | Utilized Trimble X9's high-resolution scan settings for capturing large-span structures, vertical components, and minor surface irregularities             |
| Gap Management                                | Conducted supplementary scans from alternate angles; point cloud refinement and merging handled in <b>Trimble RealWorks</b> and <b>Trimble Perspective</b> |
| Data Validation                               | Outlier filtering and noise reduction applied; <b>registration error maintained &lt;5 mm RMSE</b> for consistent dimensional reliability                   |

### Equipment Specifications:

| Feature                      | Trimble X9                                      |
|------------------------------|-------------------------------------------------|
| Scan Speed                   | 1,000,000 pts/sec                               |
| Time to Complete a Full Scan | 4-6 minutes per setup (depending on resolution) |
| Point Cloud Accuracy         | ±3 mm at 20m                                    |
| Maximum Scanning Range       | 150m                                            |
| Field Workflow               | Auto-calibration                                |
| Imaging                      | Standard HDR                                    |
| Durability                   | IP55                                            |

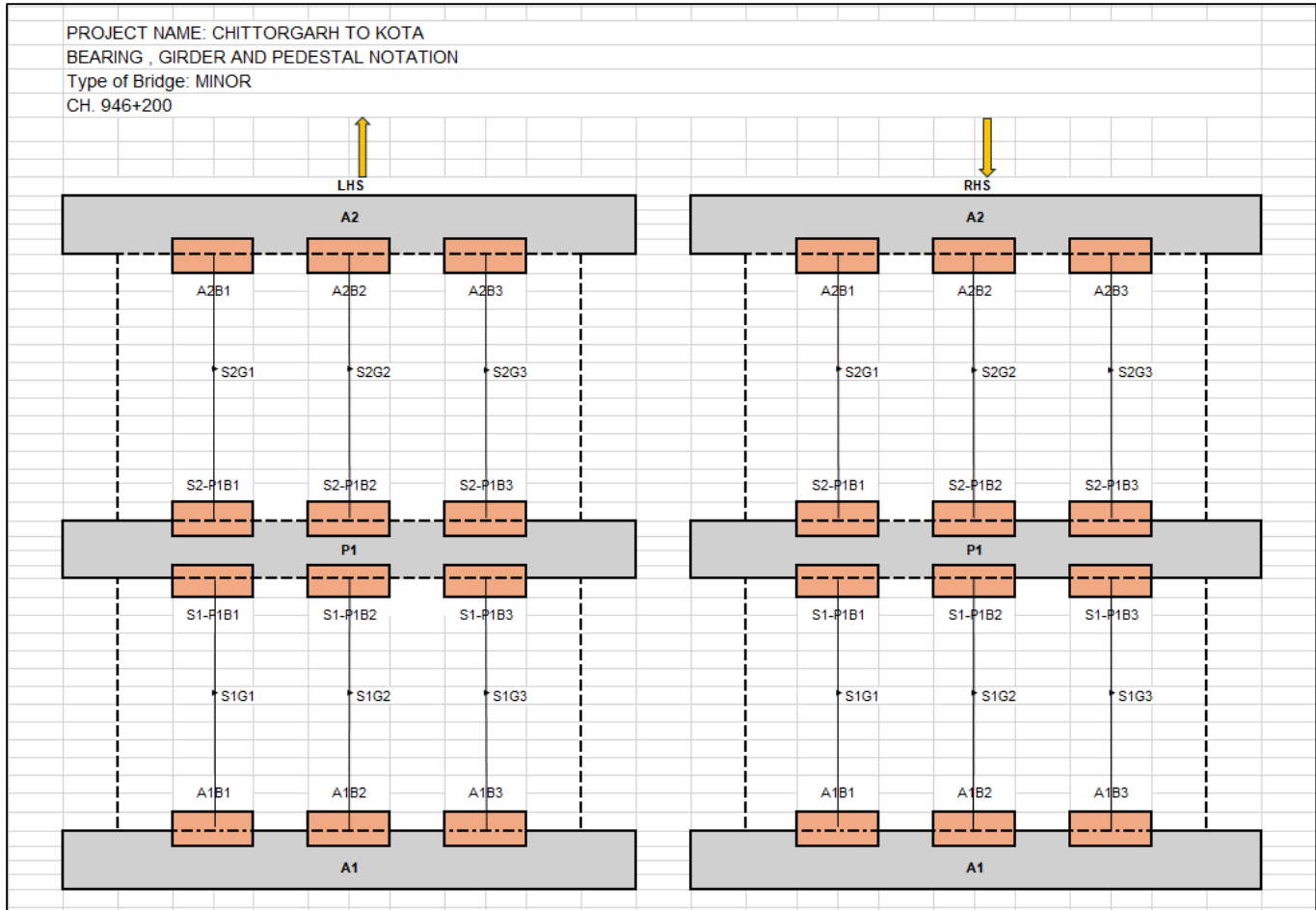
## Table of Content

| <b>Sr. No.</b> | <b>Description</b>              | <b>Page No.</b> |
|----------------|---------------------------------|-----------------|
| 0.             | Phase 1 – Scope of Deliverables | 5               |
| 1.             | Structure Layout                | 8               |
| 2.             | Layout Legends                  | 9               |
| 3.             | Distress Nomenclature           | 10              |
| 4.             | Observation Sheet               | 10              |
| 5.             | Photos                          | 15              |

## 0. Phase 1 – Scope of Deliverables

| Work Stage                     | Scope / Deliverables                                                     | Location in Report                          |
|--------------------------------|--------------------------------------------------------------------------|---------------------------------------------|
| Accuracy Setup & Scan Planning | Specification of accuracy levels, point cloud density, and scan coverage | Technical Scope of LiDAR Inspection         |
| Technical Setup                | Preparation for LiDAR equipment deployment                               | Technical Scope of LiDAR Inspection         |
| Objective Framing              | Component prioritization for scanning                                    | Structural Measurements + Observation Sheet |
| Field Data Collection          | LiDAR data acquisition                                                   | Structure Layout + Photos                   |
| Resolution Analysis            | Data density and resolution analysis                                     | Technical Scope of LiDAR Inspection         |
| Gap Filling                    | Quality enhancement of point cloud                                       | Embedded in Visual/Photo Section            |
| Point Cloud Analysis           | Structural details and interpretation                                    | Observation Sheet + Distress Photos         |

# 1. Structure Layout



## 2. Layout Legends

| S. No | Description               | Notation |
|-------|---------------------------|----------|
| 1     | Abutment                  | AB       |
| 2     | Foundation of Abutment    | AF       |
| 3     | Abutment Cap              | AC       |
| 4     | Bearing Pedestal          | BP       |
| 5     | Bearings                  | BR       |
| 6     | Foundation of Pier        | PF       |
| 7     | Pier                      | PR       |
| 8     | Pier Cap                  | PC       |
| 9     | Span                      | SP       |
| 10    | Slab span                 | SL       |
| 11    | Box Girder span           | BG       |
| 12    | I Girder                  | IG       |
| 13    | Rectangular Girder        | RG       |
| 14    | Square Girder             | SG       |
| 15    | T Beam Girders            | TB       |
| 16    | Concrete Deck             | CC       |
| 17    | Bituminous Deck           | BD       |
| 18    | Slab b/n girders          | BS       |
| 19    | Cross Girders             | CG       |
| 20    | Railing LHS               | RL       |
| 21    | Railing RHS               | RR       |
| 22    | Footpath LHS              | FL       |
| 23    | Footpath RHS              | RR       |
| 24    | Expansion Joints          | EJ       |
| 25    | Pourable/Crack Joint Seal | PJ       |
| 26    | Compression Joint Seal    | CJ       |
| 27    | Assembly Joint Seal       | AJ       |
| 28    | Strip Seal Joint          | SJ       |
| 29    | Drainage Spouts           | DS       |
| 30    | Wing Wall                 | WW       |
| 31    | Reinforced earth wall     | RE       |
| 32    | Approach Slab             | AS       |

### 3. Distress Nomenclature

| Legend | Description                               | Legend    | Description         |
|--------|-------------------------------------------|-----------|---------------------|
| BG     | Bulging                                   | SC        | Steel Corroded      |
| CRA    | Map Crack/ Crazing                        | SE        | Steel Exposed       |
| RC     | Crack along main reinforcement            | SPE       | Seepage Evident     |
| RP     | Crack perpendicular to Main Reinforcement | SPM       | Seepage Marks       |
|        |                                           | SL        | Scaling             |
| MC     | Minor Crack                               | SG/(G)(S) | Sag in Girder, Slab |
| WC     | Wide Crack                                | EFF       | Efflorescence       |
| SPL    | Spalling                                  | STL       | Settlement          |
| DL     | Delaminated Section                       | H         | Hollow Section      |
| HC     | Honey Combing                             | VEG       | Vegetation Growth   |
| TL     | Tilt                                      | PD        | Pedestal Damage     |

## 4. Observations

**LHS:**

| Element Name | Observation    | Location of Distress | Photo No. | Units | Distress Details |      |       | Area of Repair (Sq.m/R MT/m3) |
|--------------|----------------|----------------------|-----------|-------|------------------|------|-------|-------------------------------|
|              |                |                      |           |       | L(M)             | B(M) | D(M)  |                               |
| Girder 1     | Hollow Section | -                    | 1         | M     | 0.18             | 0.14 | 0.046 | 0.0011                        |
| Girder 3     | Damage         | -                    | 2         | M     | 0.38             | 0.12 | 0.036 | 0.0016                        |
| Girder 2     | Hollow section | -                    | 3         | M     | 0.36             | 0.35 | 0.034 | 0.0042                        |
| Girder 2     | Damage         | -                    | 4         | M     | 0.44             | 0.11 |       | 0.0484                        |
| Girder 1     | Hollow Section | -                    | 5         | M     | 1.51             | 0.60 | 0.046 | 0.0416                        |
| Girder 1     | Hollow Section | -                    | 6         | M     | 0.94             | 0.62 | 0.042 | 0.0244                        |
| Girder       | Damage         | -                    | 7         | M     | 0.28             | 0.12 | 0.049 | 0.00164                       |

**RHS:**

| Element Name | Observation    | Location of Distress | Photo No. | Units | Distress Details |      |       | Area of Repair (Sq./RM T) |
|--------------|----------------|----------------------|-----------|-------|------------------|------|-------|---------------------------|
|              |                |                      |           |       | L(M)             | B(M) | D(M)  |                           |
| Pier 1       | Honeycombing   | -                    | 8         | M     | 1.74             | 0.54 | -     | 0.939                     |
| Girder 3     | Hollow Section | -                    | 9         | M     | 1.28             | 0.60 | 0.047 | 0.0360                    |
| Girder 3     | Hollow Section | -                    | 10        | M     | 0.92             | 0.57 | 0.045 | 0.0235                    |
| Girder 3     | Hollow Section | -                    | 11        | M     | 0.28             | 0.14 | 0.055 | 0.0021                    |
| Girder 3     | Hollow Section | -                    | 12        | M     | 1.00             | 0.62 | 0.031 | 0.0192                    |
| Girder 3     | Damage         | -                    | 13        | M     | 0.52             | 0.19 | 0.032 | 0.0031                    |
| Girder 2     | Hollow Section | -                    | 14        | M     | 0.61             | 0.34 | 0.037 | 0.0076                    |
| Girder 2     | Hollow Section | -                    | 15        | M     | 0.28             | 0.22 | 0.033 | 0.0020                    |
| Pier Cap     | Damage         | -                    | 16        | M     | 0.17             | 0.15 | 0.019 | 0.0004                    |
| Girder 3     | Hollow Section | -                    | 17        | M     | 0.60             | 0.22 | 0.018 | 0.0023                    |
| Girder 2     | Hollow Section | -                    | 18        | M     | 0.45             | 0.40 | 0.023 | 0.0041                    |
| Pier Cap     | honeycombing   | -                    | 19        | M     | 0.27             | 0.09 | -     | 0.024                     |

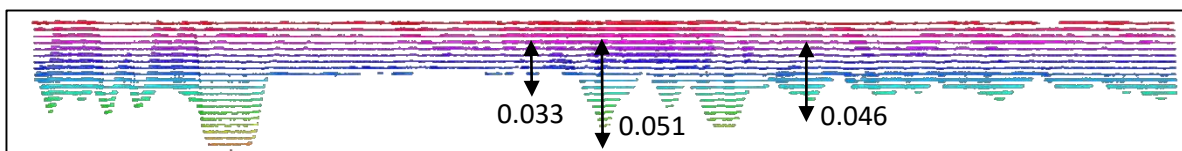
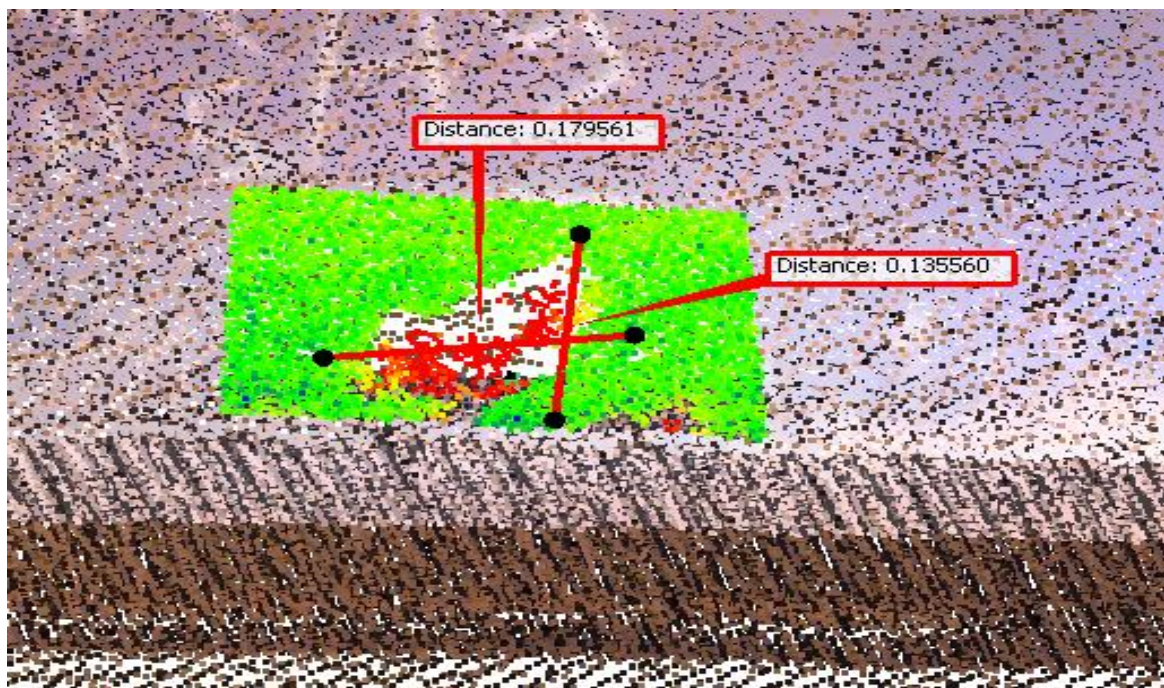
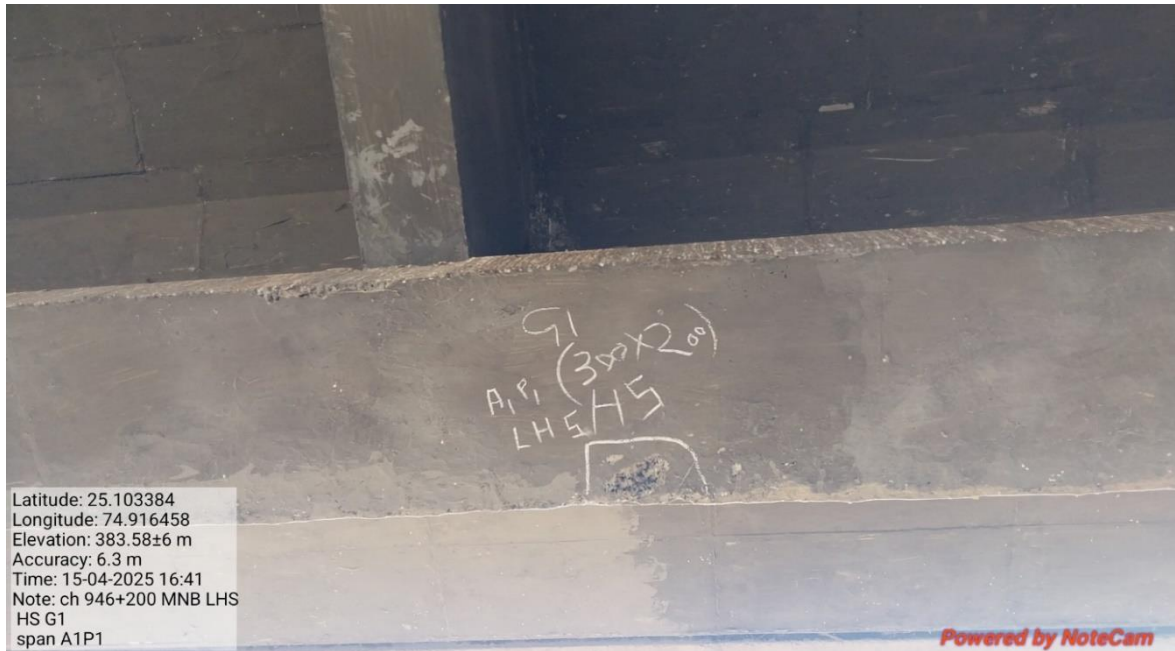
### Observation sheet Non- Structural Elements

| Sl. No. | Description                           | Distress Notation | Observation       | Location |       | Distress Details |       |       | Area of Repair |
|---------|---------------------------------------|-------------------|-------------------|----------|-------|------------------|-------|-------|----------------|
|         |                                       |                   |                   | X (m)    | Y (m) | L (m)            | W (m) | D (m) | Sqm            |
| 1       | Return wall                           |                   | OK                |          |       |                  |       |       |                |
| 2       | Riding return wall                    |                   | NA                |          |       |                  |       |       |                |
| 3       | Wing wall                             |                   | NA                |          |       |                  |       |       |                |
| 4       | Butterfly walls at top of median wall |                   | NA                |          |       |                  |       |       |                |
| 5       | Dirt wall at top of abutment          |                   | Ok                |          |       |                  |       |       |                |
| 6       | Gabion walls                          |                   | NA                |          |       |                  |       |       |                |
| 7       | Hazard Marker sign board              |                   | OK                |          |       |                  |       |       |                |
| 8       | Hand railing                          |                   | NA                |          |       |                  |       |       |                |
| 9       | Crash barrier                         |                   | Damaged (LHS/RHS) |          |       |                  |       |       |                |
| 10      | Median wall                           |                   | NA                |          |       |                  |       |       |                |
| 11      | Pipe Hand Railing (on crash Barrier)  |                   | OK                |          |       |                  |       |       |                |
| 12      | Cleaning of expansion joints          |                   | Choked            |          |       |                  |       |       |                |
| 13      | Wheel guard (Mounted & Unmounted)     |                   | NA                |          |       |                  |       |       |                |
| 14      | Footpath                              |                   | Damaged (RHS)     |          |       |                  |       |       |                |
| 15      | Drainage Spouts                       |                   | Choked            |          |       |                  |       |       |                |
| 16      | Drainage Spout Grating                |                   | Choked            |          |       |                  |       |       |                |
| 17      | Drainage Spout Extension Pipe         |                   | OK                |          |       |                  |       |       |                |
| 18      | Weep Holes                            |                   | NA                |          |       |                  |       |       |                |
| 19      | Fascia Panel                          |                   | NA                |          |       |                  |       |       |                |

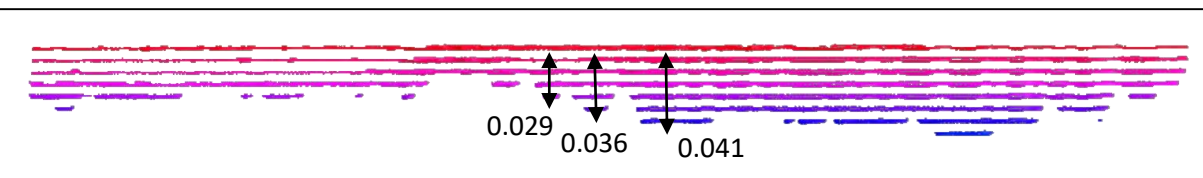
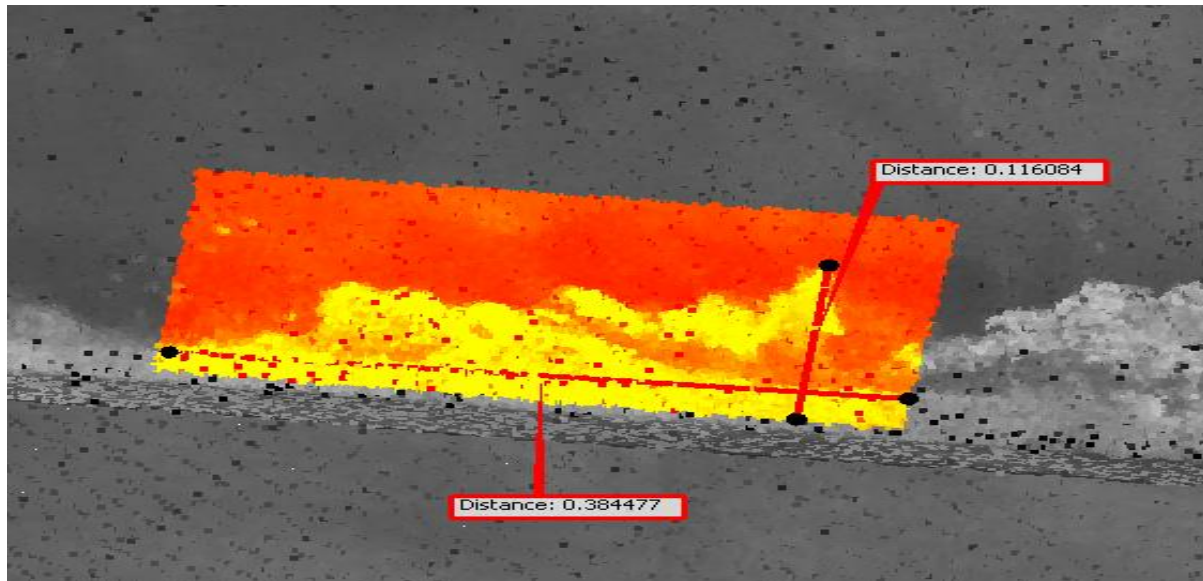
| Sl. No. | Description                      | Distress Notation | Observation | Location |       | Distress Details |       |       | Area of Repair |
|---------|----------------------------------|-------------------|-------------|----------|-------|------------------|-------|-------|----------------|
|         |                                  |                   |             | X (m)    | Y (m) | L (m)            | W (m) | D (m) | Sqm            |
| 20      | Friction Slab                    |                   | Ok          |          |       |                  |       |       |                |
| 21      | Approach slab including PCC      |                   | OK          |          |       |                  |       |       |                |
| 22      | Backfilling below Approach slab  |                   | OK          |          |       |                  |       |       |                |
| 23      | Stone Pitching in quarter coning |                   | NA          |          |       |                  |       |       |                |
| 24      | Filling in quarter coning        |                   | NA          |          |       |                  |       |       |                |
| 25      | Rigid apron                      |                   | NA          |          |       |                  |       |       |                |
| 26      | Curtain wall                     |                   | NA          |          |       |                  |       |       |                |
| 27      | Flexible apron                   |                   | NA          |          |       |                  |       |       |                |
| 28      | Vent clearance                   |                   | NA          |          |       |                  |       |       |                |
| 29      | Toe wall of Quarter coning       |                   | NA          |          |       |                  |       |       |                |
| 30      | RCC drains in tunnel             |                   | NA          |          |       |                  |       |       |                |
| 31      | Cover slab of RCC drains         |                   | NA          |          |       |                  |       |       |                |
| 32      | RCC portal of tunnel             |                   | NA          |          |       |                  |       |       |                |
| 33      | Counter fort wall                |                   | NA          |          |       |                  |       |       |                |
| 34      | Booths structure                 |                   | NA          |          |       |                  |       |       |                |
| 35      | RCC Booth guard                  |                   | NA          |          |       |                  |       |       |                |
| 36      | C.C Panels including PCC         |                   | NA          |          |       |                  |       |       |                |
| 37      | Pipe drain                       |                   | NA          |          |       |                  |       |       |                |
| 38      | Pipe drain cover                 |                   | NA          |          |       |                  |       |       |                |
| 39      | Drain openings                   |                   | NA          |          |       |                  |       |       |                |

## **CH: 946+200 – MNB**

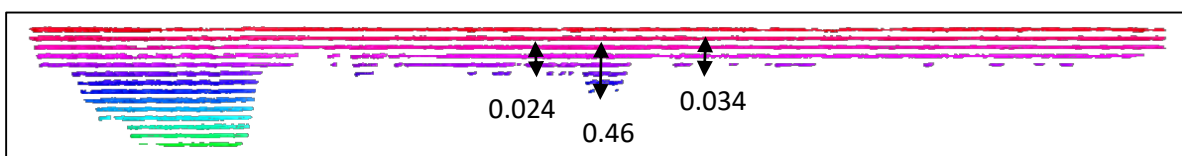
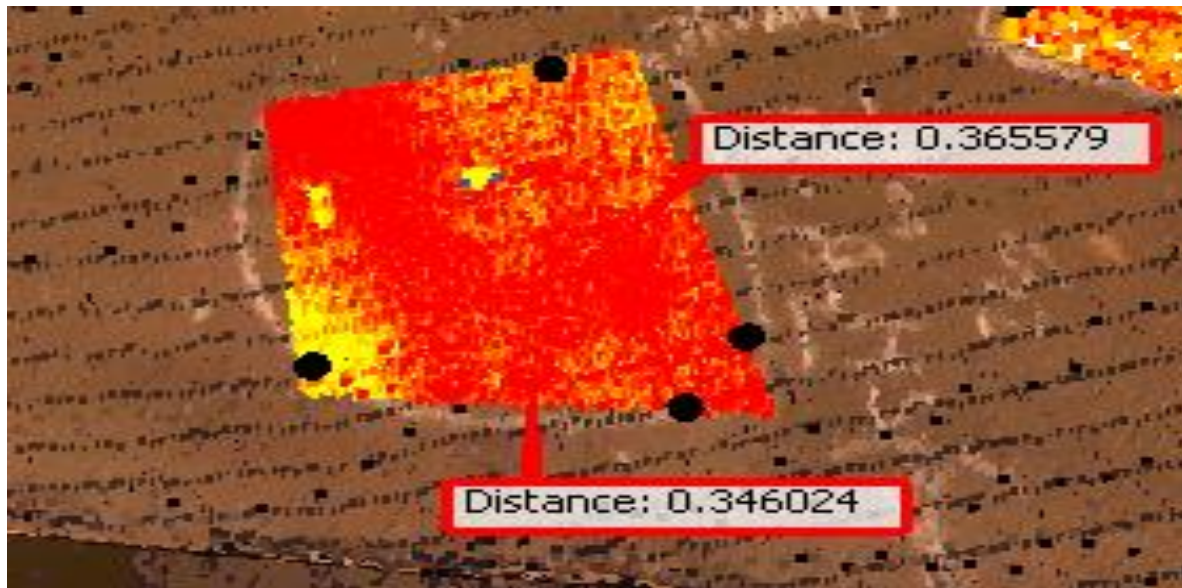
DISTRESS PHOTOS



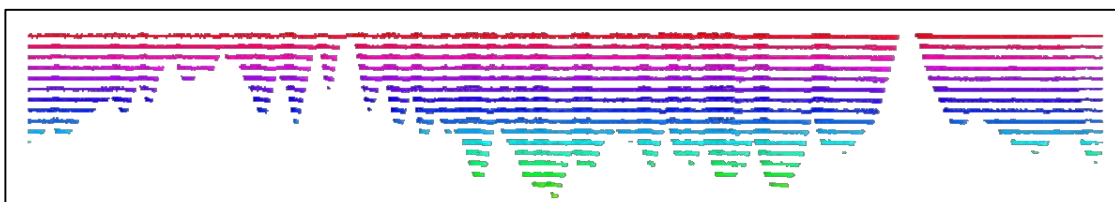
**1. Hollow section observed on Girder 1 LHS, span A1–P1, measuring 0.18 m in length and 0.14 m in width (area: 0.0252 m<sup>2</sup>). Depth ranges from 0.033 m (min) to 0.051 m (max), with an average depth of 0.046 m. The calculated volume of the distress is 0.0011 m<sup>3</sup>.**



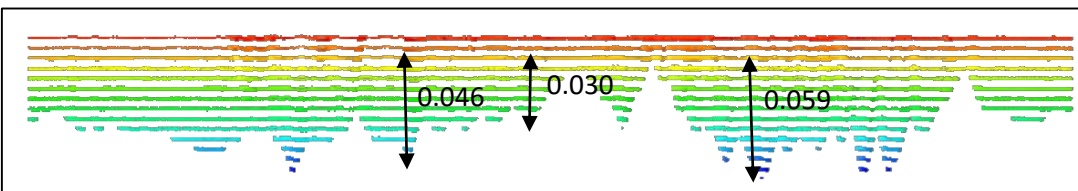
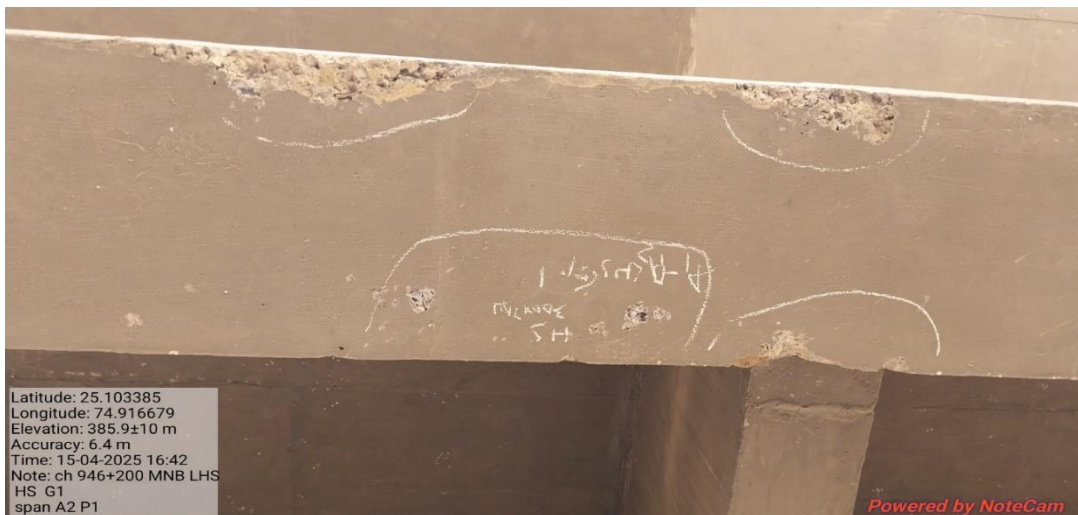
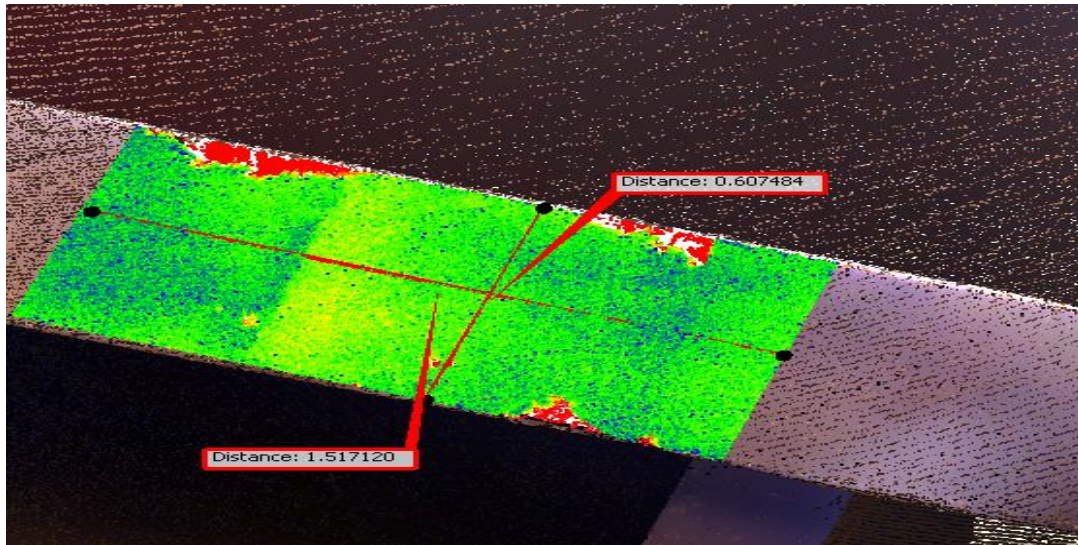
2. **Damage observed on Girder 3, span A2–P1, measuring 0.38 m in length and 0.12 m in width. Depth ranges from 0.029 m (min) to 0.041 m (max), with an average depth of 0.036 m. The calculated volume of the distress is 0.00164 m<sup>3</sup>.**



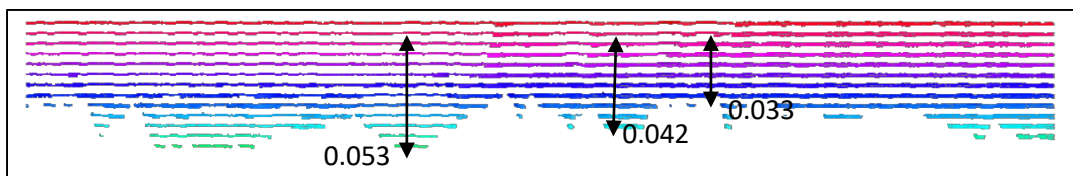
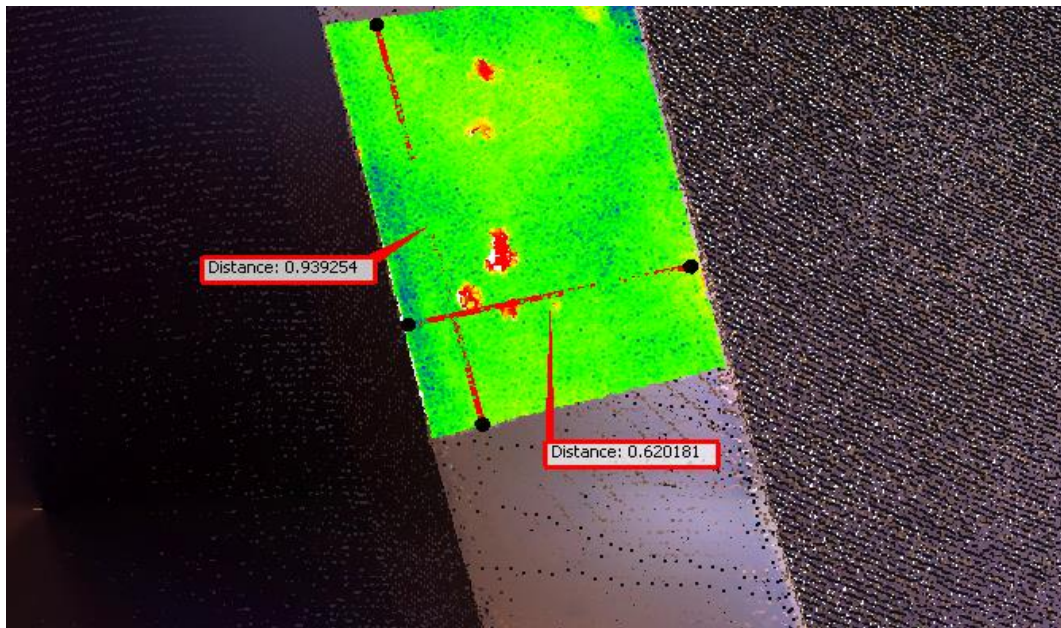
**3. Hollow section observed on Girder 2, span A2–P1, measuring 0.36 m in length and 0.35 m in width. Depth ranges from 0.024 m (min) to 0.046 m (max), with an average depth of 0.034 m. The calculated volume of the distress is 0.0042 m<sup>3</sup>.**



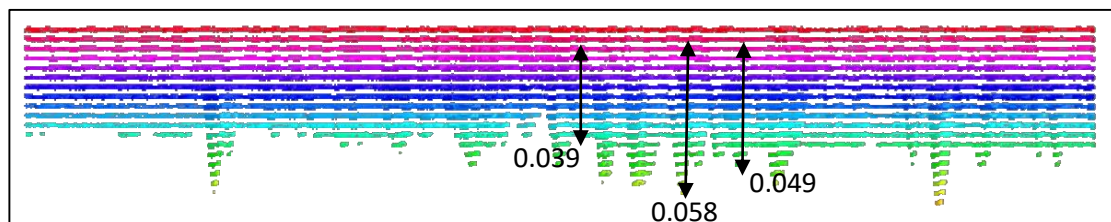
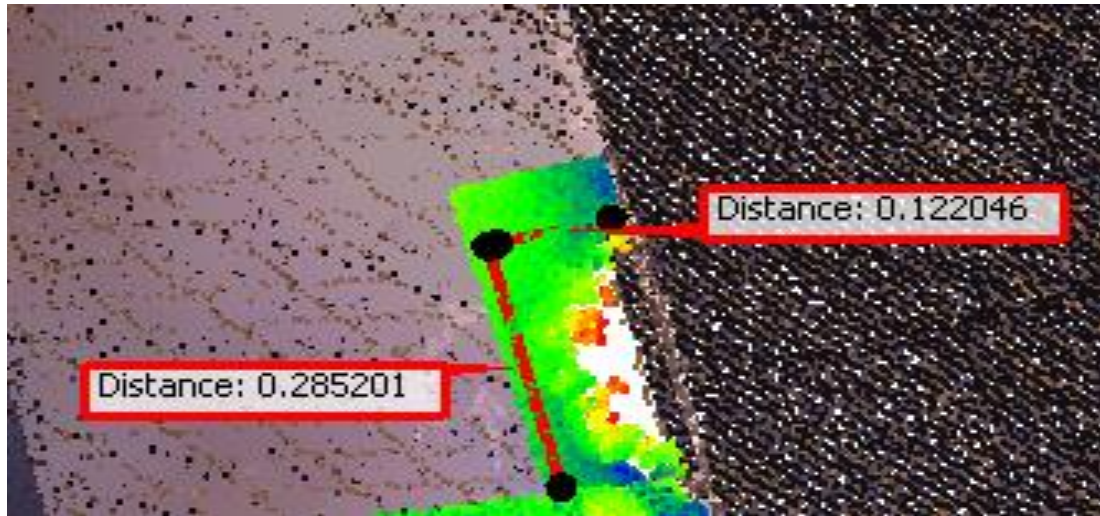
**4. Damage observed on girder 2 span A2P1 measuring 0.44 m in length and 0.11 m in width, covering an area of 0.0484 m<sup>2</sup>.**



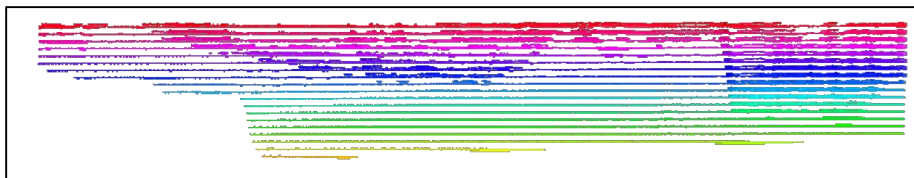
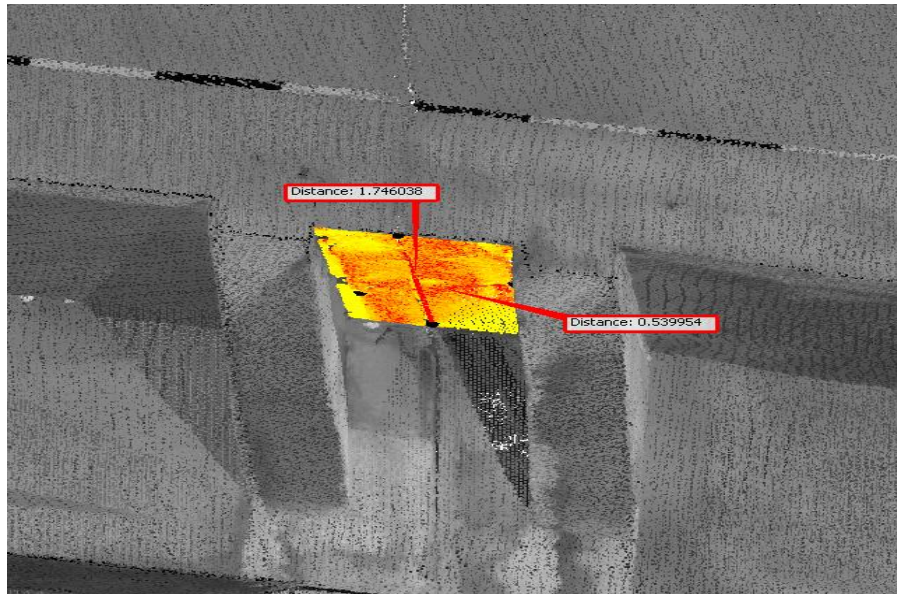
**5. Hollow section on Girder 1 span A2–P1, measuring 1.51 m in length and 0.60 m in width. Depth ranges from 0.030 m (min) to 0.059 m (max), with an average depth of 0.046 m. The calculated volume of the distress is 0.0416 m<sup>3</sup>.**



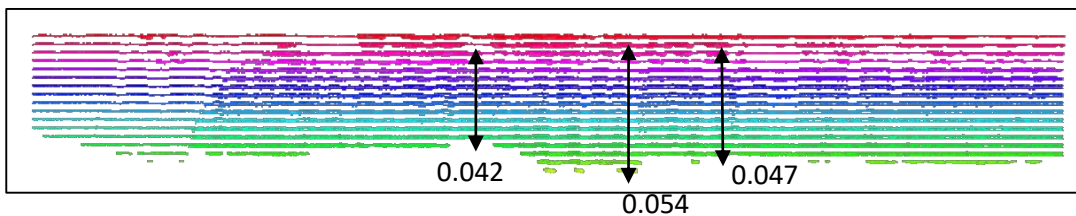
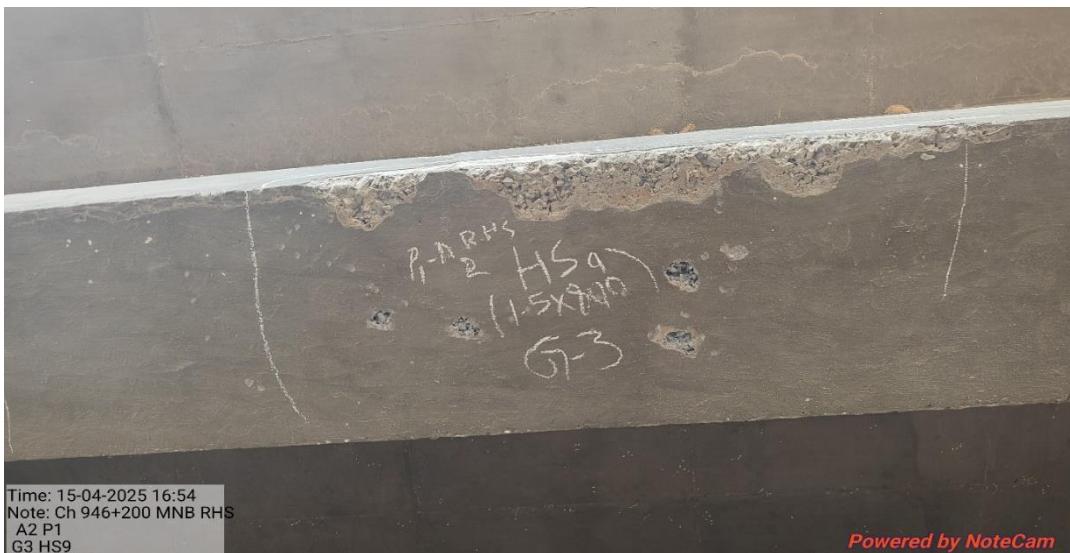
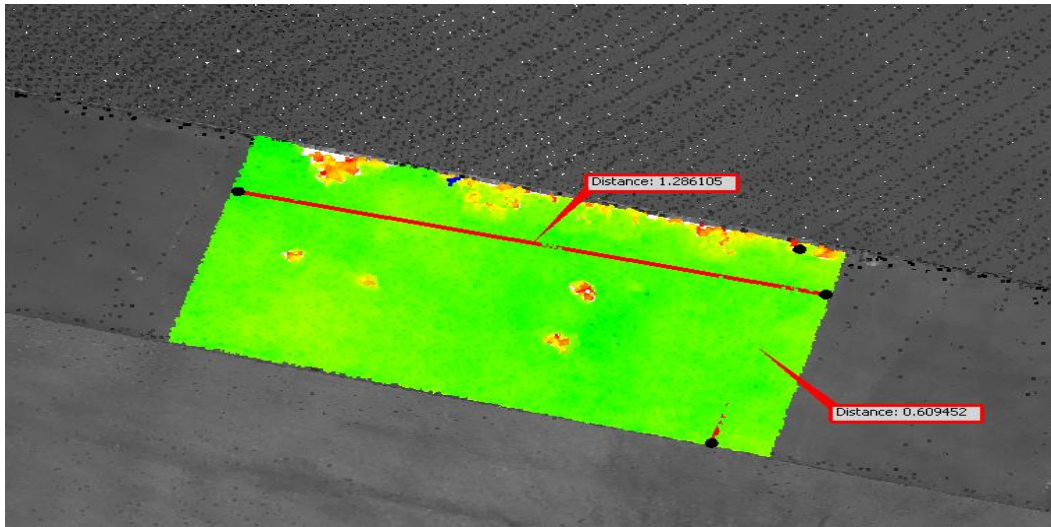
**6. Hollow section on A2–P1 Girder 1, measuring 0.94 m in length and 0.62 m in width. Depth ranges from 0.033 m (min) to 0.053 m (max), with an average depth of 0.042 m. The calculated volume of the distress is 0.0244 m<sup>3</sup>.**



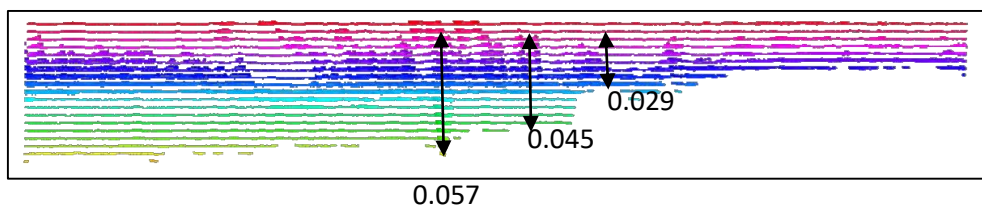
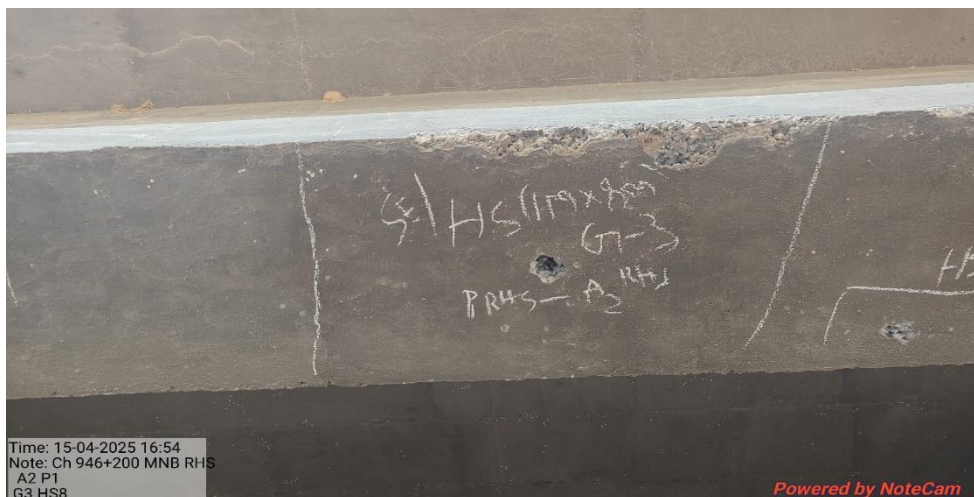
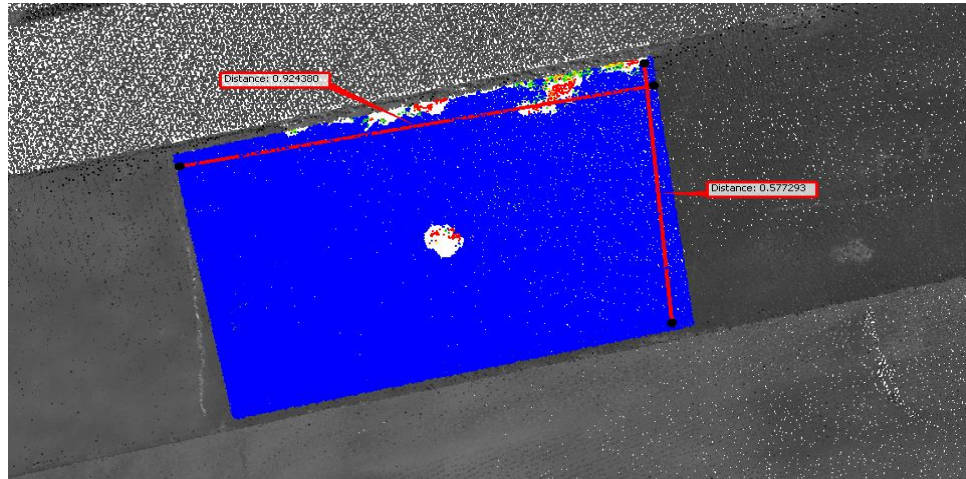
**7. Damage observed on A2–P1 Girder 1 measuring 0.28 m in length and 0.12 m in width. Depth ranges from 0.039 m (min) to 0.058 m (max), with an average depth of 0.049 m. The calculated volume of the distress is 0.0164 m<sup>3</sup>.**



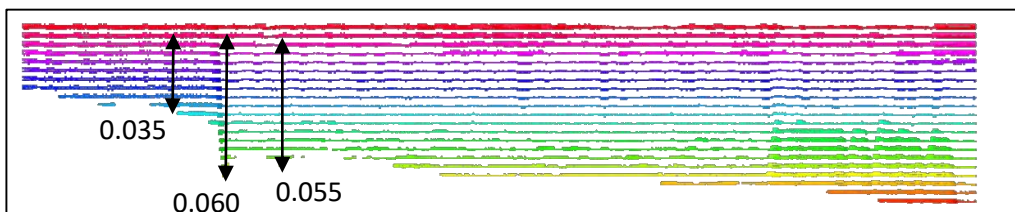
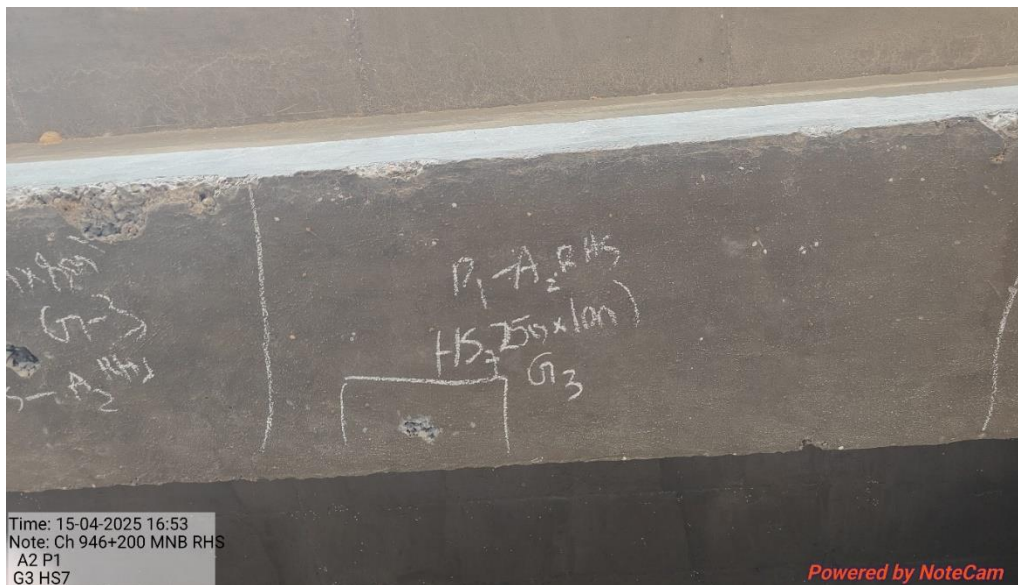
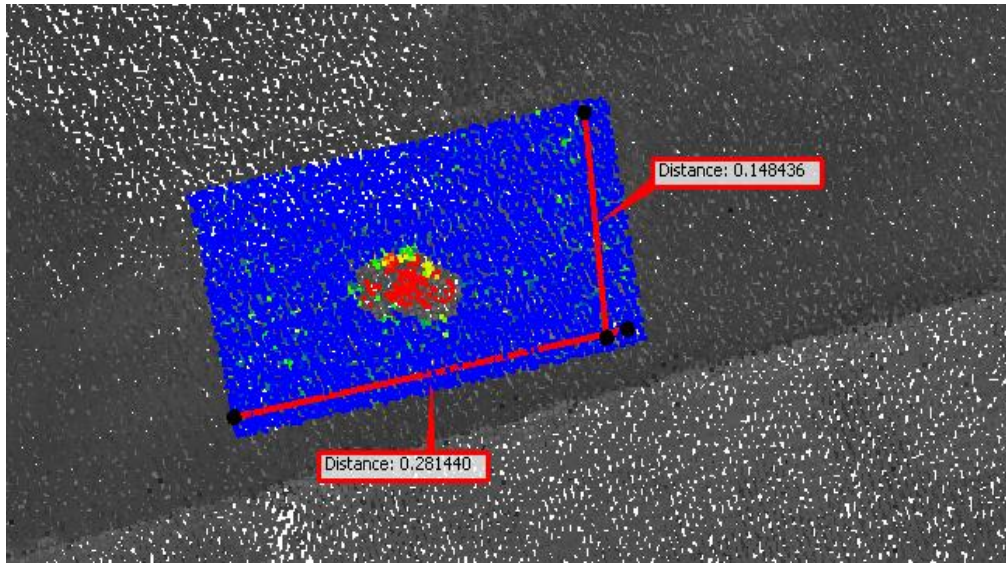
**8. Honeycombing observed on P1 median side, measuring 1.74 m in length and 0.54 m in width, covering an area of 0.9396 m<sup>2</sup>.**



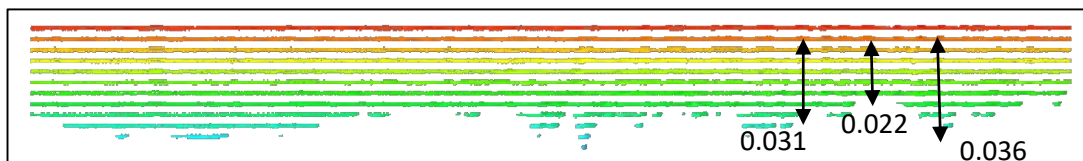
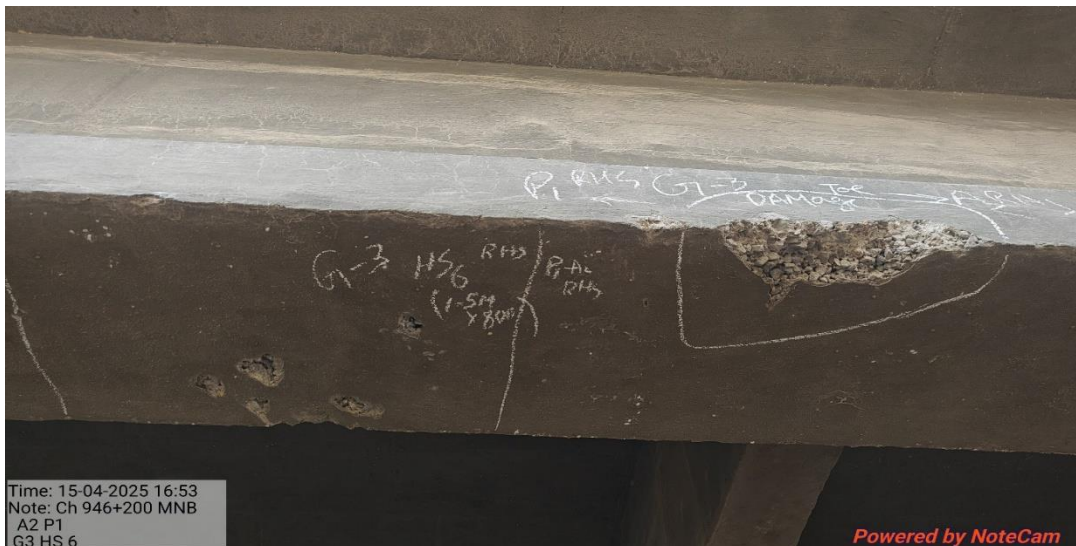
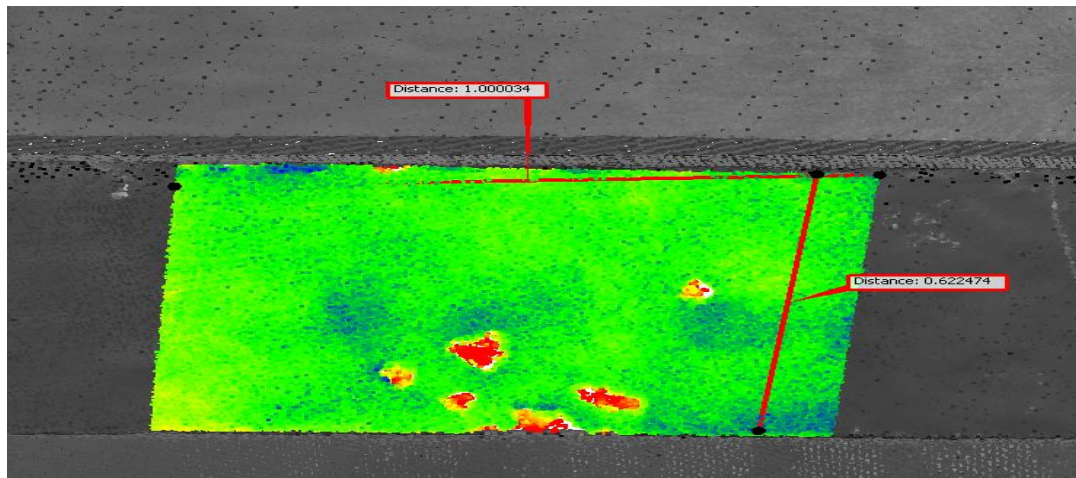
**9. Hollow section on A2–P1 Girder 3 HS9, measuring 1.28 m in length and 0.60 m in width. Depth ranges from 0.042 m (min) to 0.054 m (max), with an average depth of 0.047 m. The calculated volume of the distress is 0.360 m<sup>3</sup>.**



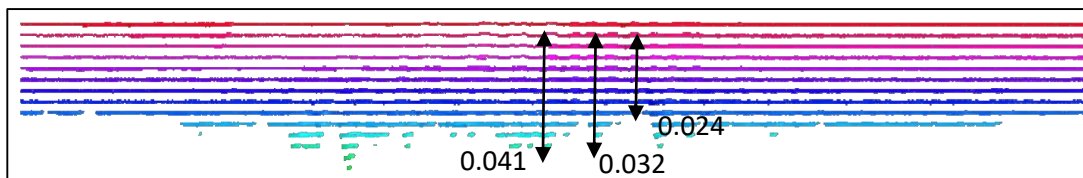
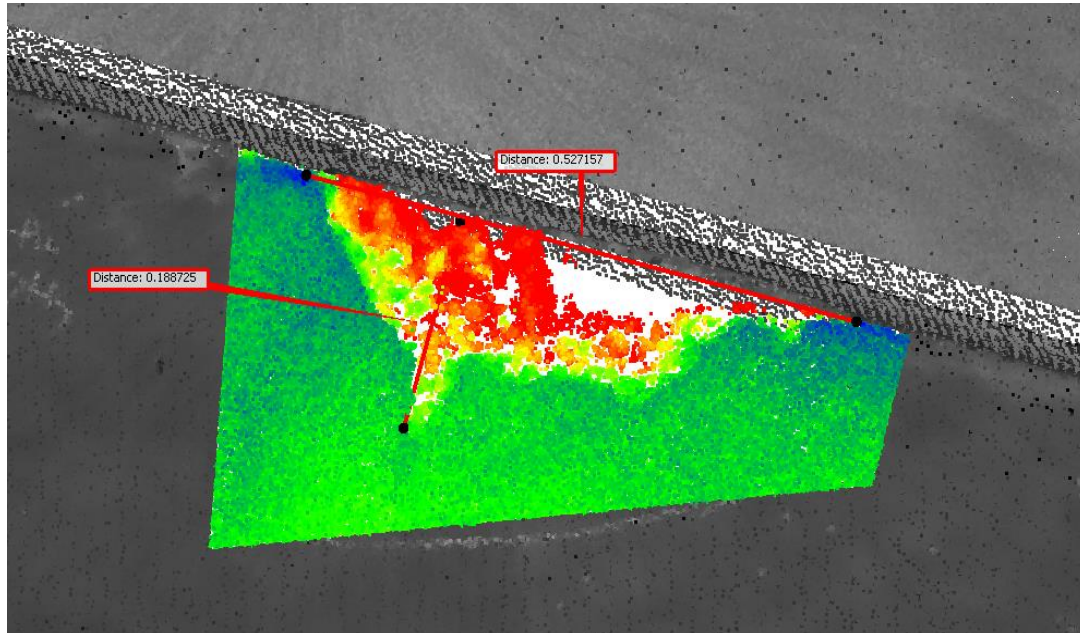
**10. Hollow section on A2–P1 Girder 3 HS8, measuring 0.92 m in length and 0.57 m in width. Depth ranges from 0.029 m (min) to 0.057 m (max), with an average depth of 0.045 m. The calculated volume of the distress is 0.0235 m<sup>3</sup>.**



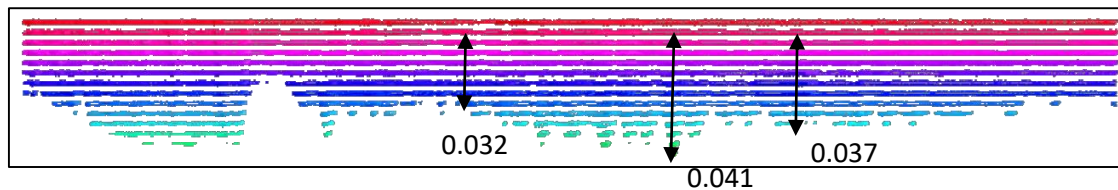
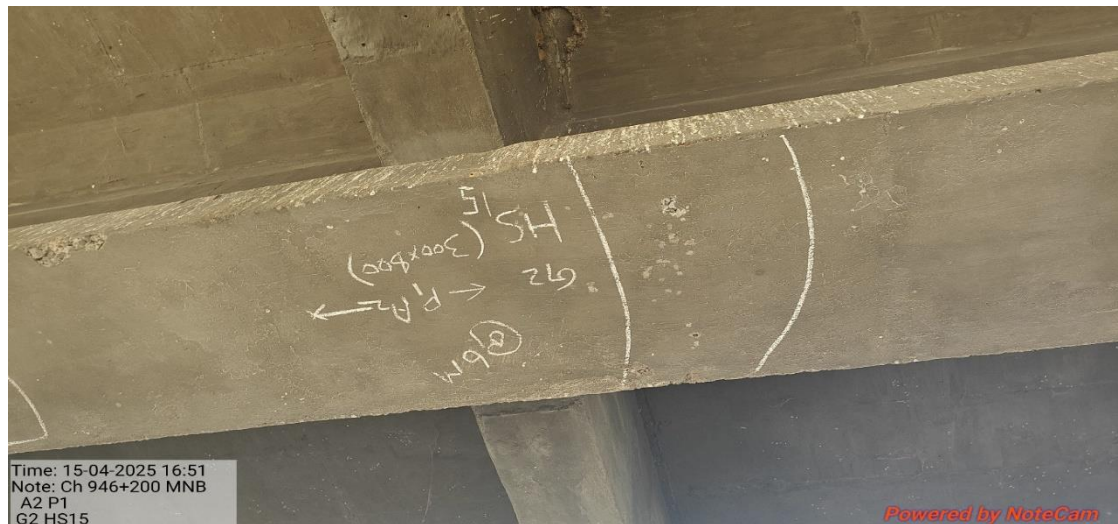
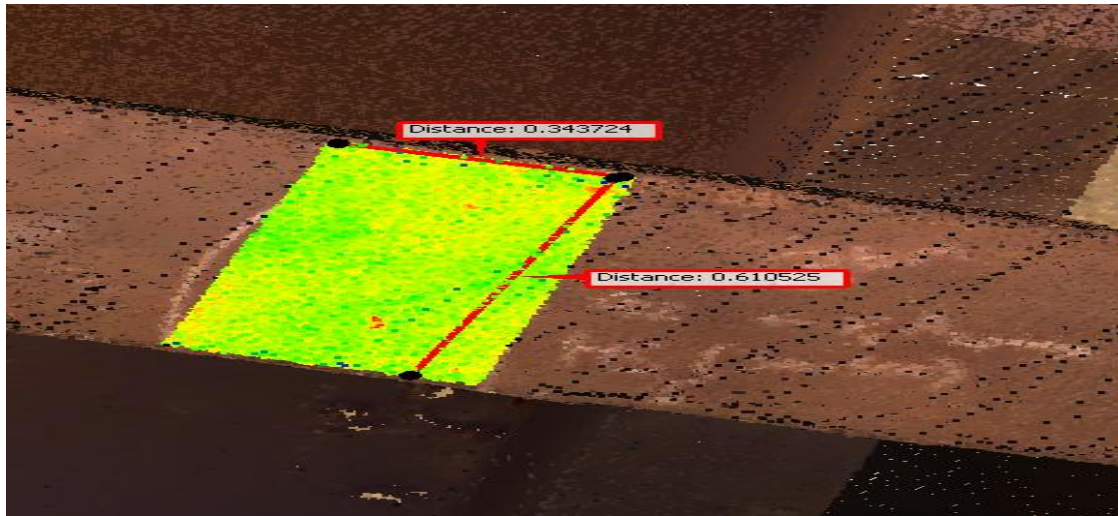
**11. Hollow section on A2–P1 Girder 3 HS7, measuring 0.28 m in length and 0.14 m in width. Depth ranges from 0.035 m (min) to 0.060 m (max), with an average depth of 0.055 m. The calculated volume of the distress is 0.0215 m<sup>3</sup>.**



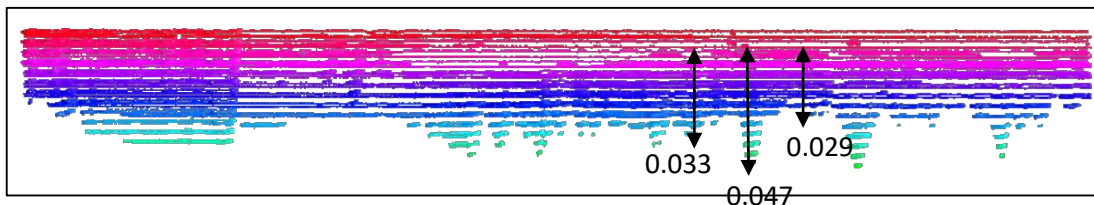
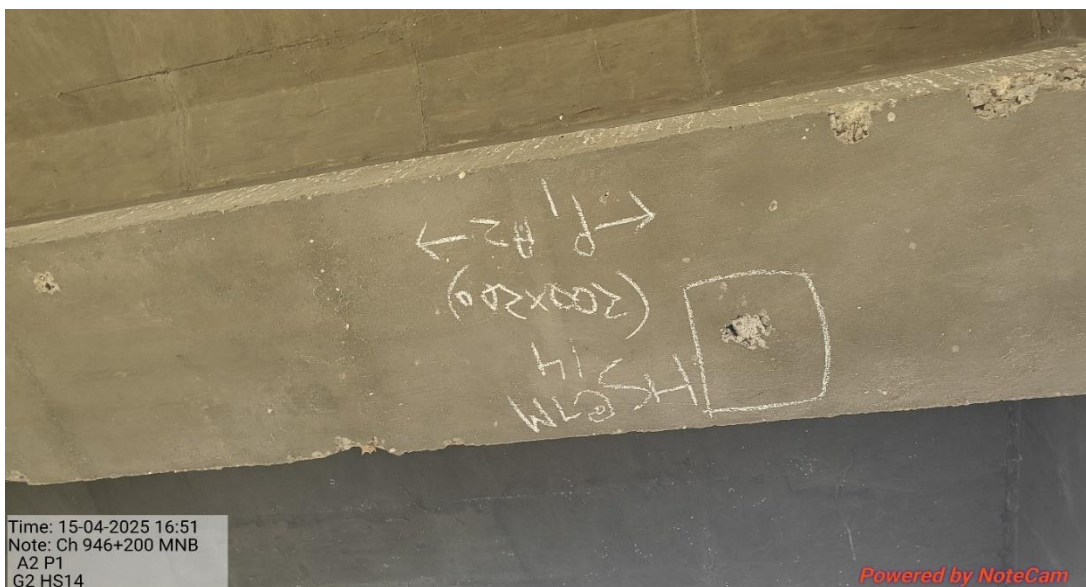
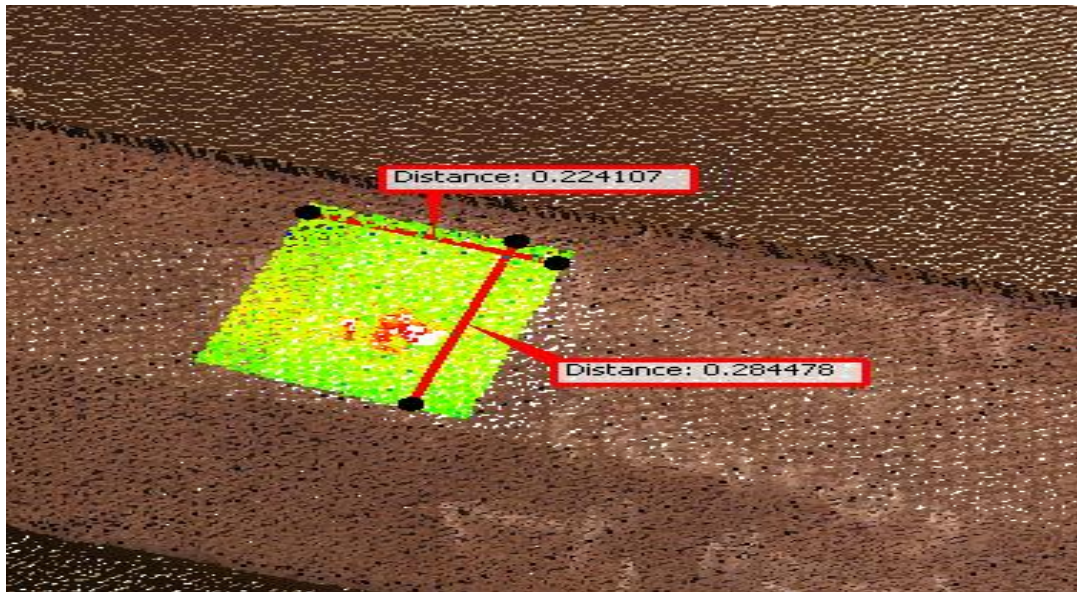
**12. Hollow section on A2-P1 Girder 3 HS6, measuring 1.00 m in length and 0.62 m in width. Depth ranges from 0.022 m (min) to 0.036 m (max), with an average depth of 0.031 m. The calculated volume of the distress is 0.0868 m<sup>3</sup>.**



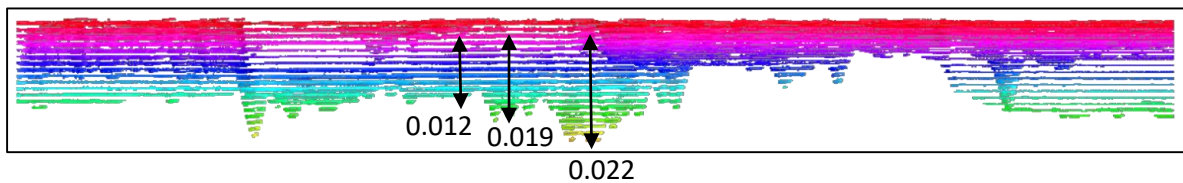
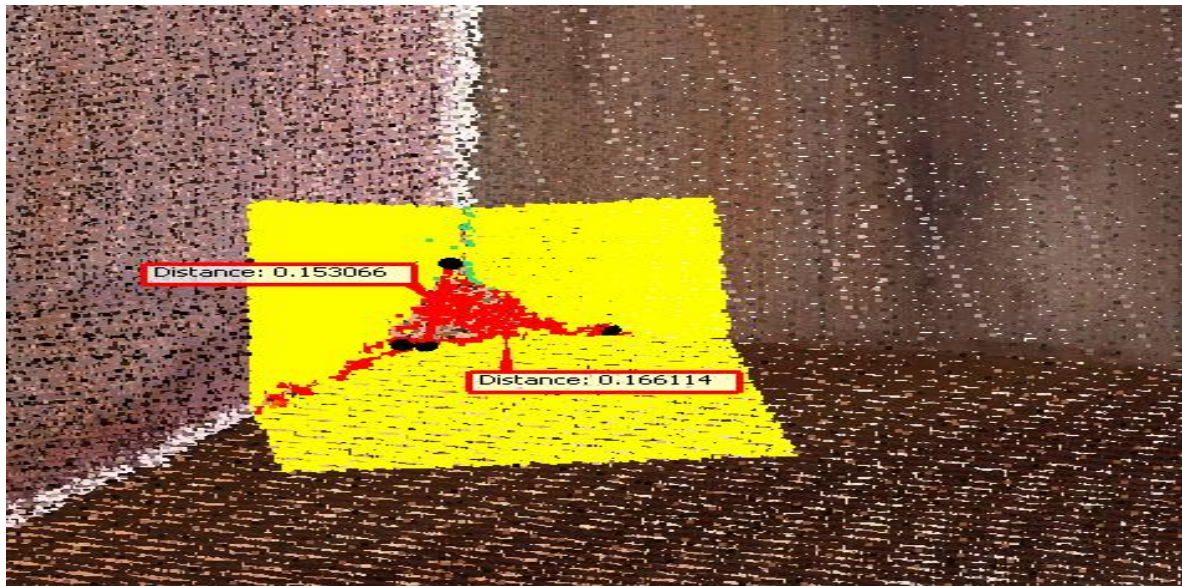
**13. Damage observed on A2–P1 Girder 3, measuring 0.52 m in length and 0.19 m in width. Depth ranges from 0.024 m (min) to 0.041 m (max), with an average depth of 0.032 m. The calculated volume of the distress is 0.0177 m<sup>3</sup>.**



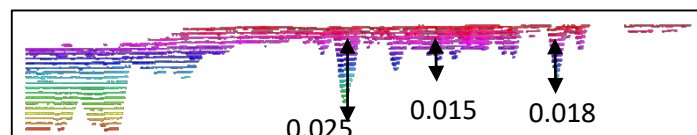
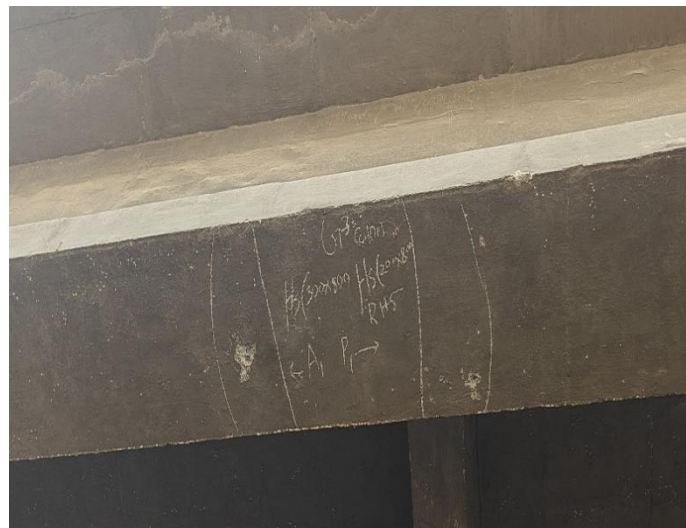
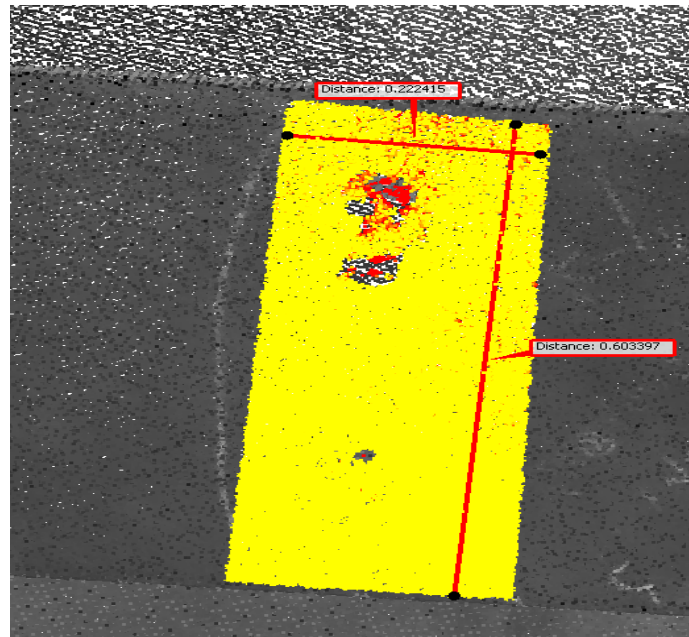
**14. Hollow section on A2–P1 Girder 2 HS15, measuring 0.61 m in length and 0.34 m in width. Depth ranges from 0.032 m (min) to 0.041 m (max), with an average depth of 0.037 m. The calculated volume of the distress is 0.0290 m<sup>3</sup>.**



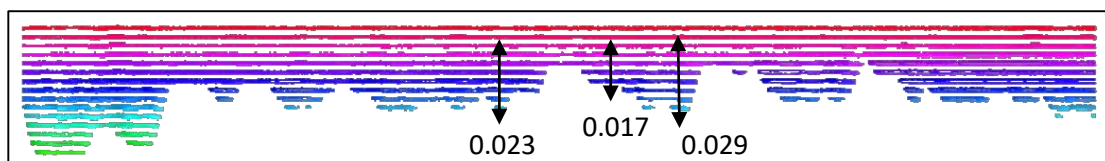
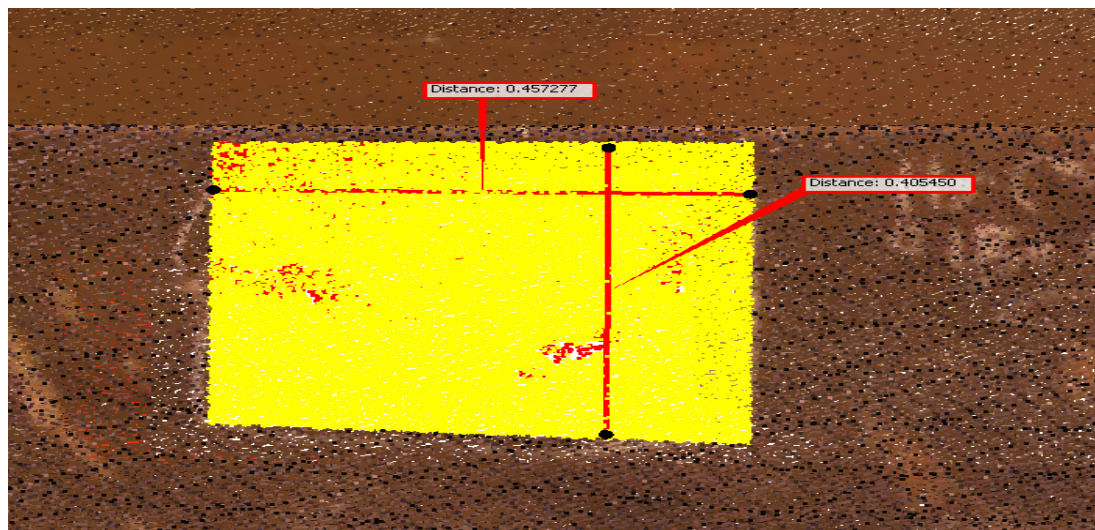
**15. Hollow section on A2–P1 Girder 2 HS14, measuring 0.28 m in length and 0.22 m in width. Depth ranges from 0.029 m (min) to 0.047 m (max), with an average depth of 0.033 m. The calculated volume of the distress is 0.00200 m<sup>3</sup>.**



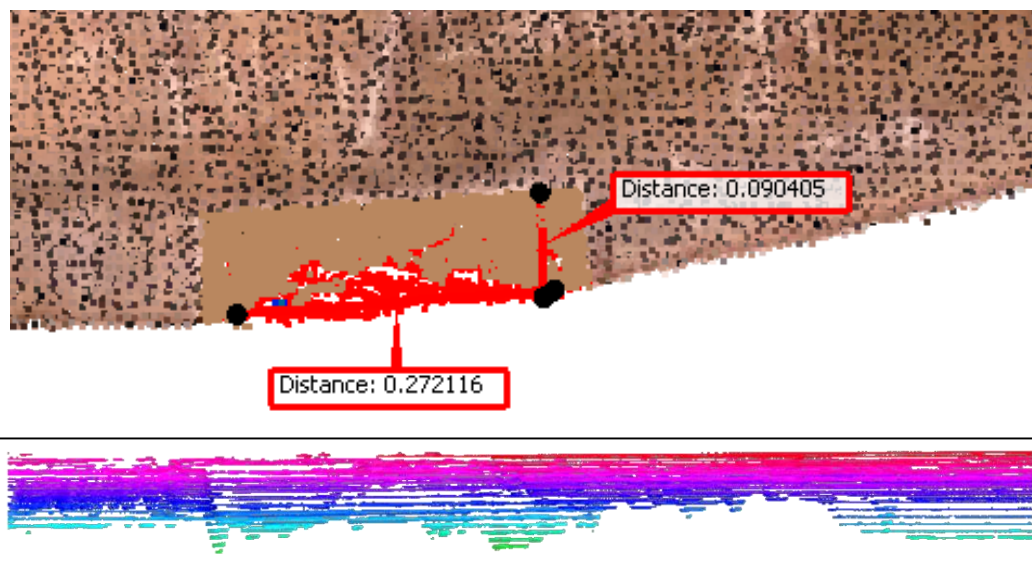
**16. Honeycombing observed on A1-P1 span pier cap bottom HC1, measuring 0.17 m in length and 0.15 m in width. Depth ranges from 0.012 m (min) to 0.022 m (max), with an average depth of 0.019 m. The calculated volume of the distress is 0.0004 m<sup>3</sup>.**



**17. Hollow section on A1–P1 span Girder 3, measuring 0.60 m in length and 0.22 m in width. Depth ranges from 0.015 m (min) to 0.025 m (max), with an average depth of 0.018 m. The calculated volume of the distress is 0.0023 m<sup>3</sup>.**



**18. Hollow section on A1–P1 span Girder 2, measuring 0.45 m in length and 0.40 m in width. Depth ranges from 0.017 m (min) to 0.029 m (max), with an average depth of 0.023 m. The calculated volume of the distress is 0.0126 m<sup>3</sup>.**



**19. Honeycombing observed on A1–P1 span pier cap bottom (HC1), measuring 0.27 m in length and 0.09 m in width, covering an area of 0.0243 m<sup>2</sup>.**